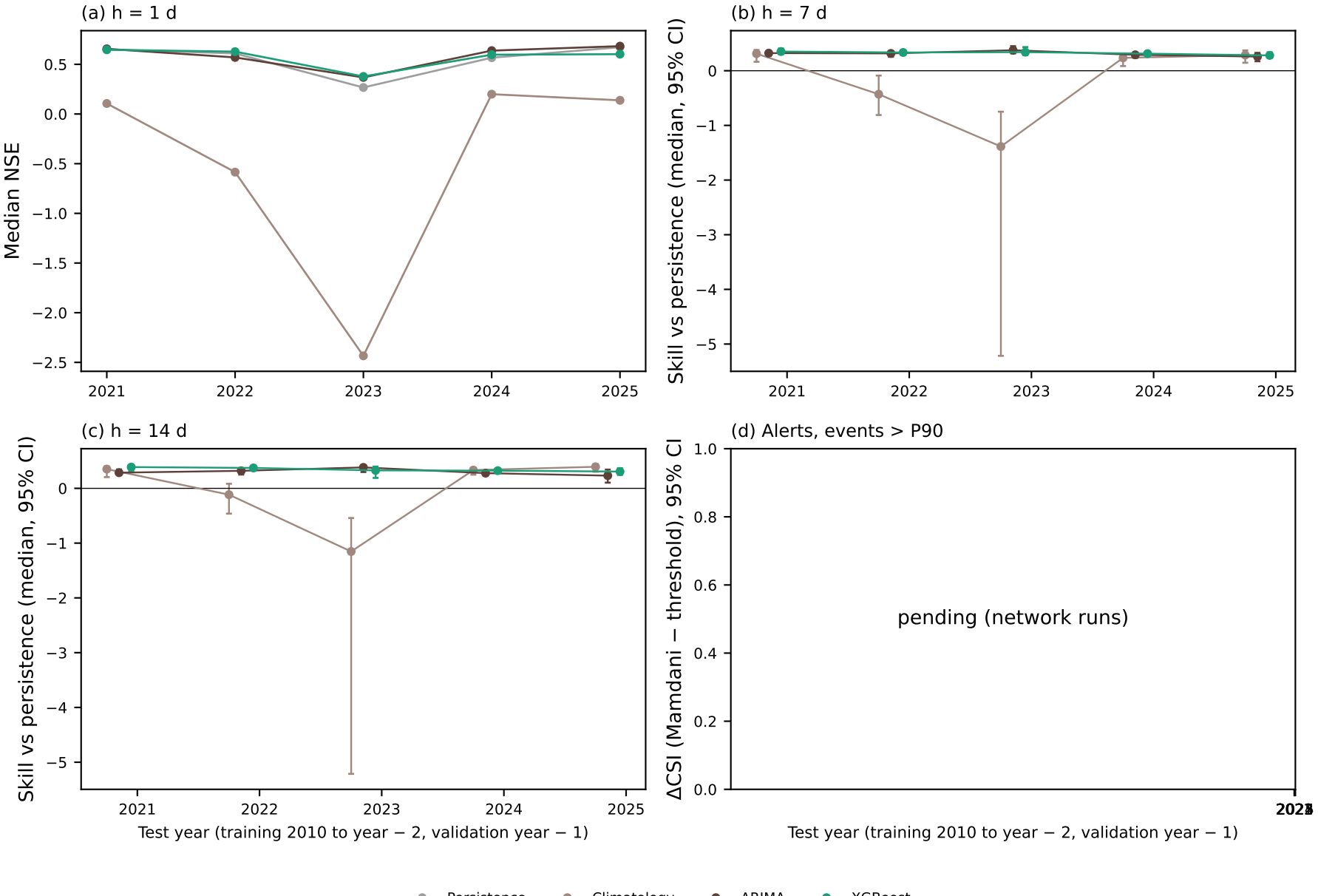
(a) h = 1 d
Median NSE
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
2021
2022
2023
2024
2025
(b) h = 7 d
Skill vs persistence (median, 95% CI)
0
−1
−2
−3
−4
−5
2021
2022
2023
2024
2025
(c) h = 14 d
Skill vs persistence (median, 95% CI)
0
−1
−2
−3
−4
−5
2021
2022
2023
2024
2025
Test year (training 2010 to year − 2, validation year − 1)
(d) Alerts, events > P90
ΔCSI (Mamdani − threshold), 95% CI
1.0
0.8
0.6
0.4
0.2
0.0
pending (network runs)
Test year (training 2010 to year − 2, validation year − 1)
Persistence
Climatology
ARIMA
XGBoost